Self-assessment Worksheet (Worksheet 6)

Do I understand this weeks material?

Real Analysis Topics	Tick the box when you feel confident with each topic
• Understand the definition of a derivative and be able to prove the derivative of a function at a point.	☐
• Be able to apply the mean value theorem, interior value theorem and Rolle's theorem.	☐
• Be able to find the radius of convergence of a given power series.	☐
• Be able to find the coefficients of a convergent power series via Taylor's theorem.	☐
• Be able to utilise Lagrange's theorem to prove specific functions are analytic.	☐

Exercises and Questions

Unless stated otherwise, you may assume any results proven in lecture.

Exercises and questions are organised and classified into three groups:

- **Starter Questions** - these questions will help cement the basic techniques.
- **Challenge Questions** - these questions are more challenging, designed to test your deeper understanding of the topic.
- **Extension Questions** - questions to try yourself in your own time to extend your knowledge and understanding.

Starter Qs

Have a go at these "warm-up" questions to get you started.

Question 1 I can do this question ☐

- Prove the the following functions are differentiable on their domains.
  (a) $f : (0, \infty) \to \mathbb{R}$ given by $f(x) = \sqrt{x+3}$.
  (b) $g : (0, \infty) \to \mathbb{R}$ given by $g(x) = \dfrac{1}{x}$.
  (c) $h : (0, \infty) \to \mathbb{R}$ given by $h(x) = 4x + \dfrac{1}{x^2}$.

Question 2 I can do this question ☐

- Suppose that $\cos(x) = \sum_{k=0}^{\infty} a_k x^k$. By using Taylor's theorem, find an explicit formula for the coefficients $a_k \in \mathbb{R}$.

Question 3 I can do this question ☐

- Using the ratio test, find the radius of convergence of the following power series
  (a) $\sum_{k=0}^{\infty} (-1)^k x^k$.
  (b) $\sum_{k=1}^{\infty} \dfrac{(-1)^{k+1} x^k}{k}$.

Question 4 I can do this question ☐

- Given that
$$\arctan(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}$$
for $|x| \leq 1$, find a power series expression for $1/(1+x^2)$ and its corresponding radius of convergence.

Question 5 I can do this question ☐

- Write down the Taylor polynomial of order 4 the function $f : (-\pi/2, \pi/2) \to \mathbb{R}$ given by $f(x) = \sec(x)$.

Challenge Qs

Now you are feeling more confident try these more challenging questions.

Question 6 I can do this question ☐

- Let $f_\alpha : \mathbb{R} \to \mathbb{R}$ given by

$$f_\alpha(x) = \begin{cases} x^\alpha \sin(1/x) & : x \neq 0 \\ 0 & : x = 0 \end{cases}$$

where $\alpha \in \mathbb{R}$. Find a value of α such that

(a) f_α is differentiable on $\mathbb{R}$ but such that f'_α is unbounded on $[0, 1]$.

(b) f_α is differentiable on $\mathbb{R}$ with f'_α continuous but not differentiable at zero.

(c) f_α is differentiable on $\mathbb{R}$ and f'_α is differentiable on $\mathbb{R}$, but such that f''_α is not continuous at zero.

Question 7 I can do this question ☐

- Prove Theorem 3.3 (ii) from Chapter 5 of the notes.

Question 8 I can do this question ☐

- Prove theorem 3.6 (Quotient Rule) from Chapter 5 of the notes.

Question 9 I can do this question ☐

- Finish the proof of Theorem 4.1 (Interior Exremum Theorem) from Chapter 5 of the notes.

Question 10 I can do this question ☐

- Prove Theorem 6.1 from Chapter 5 of the notes.

Question 11 I can do this question ☐

- Finish the proof of Theorem 7.4 (Lagrange's Remainder Theorem) from Chapter 5 of the notes.

Question 12 I can do this question ☐

- Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = \cos(x)$. Using Lagrange's remainder theorem, prove that

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}.$$

Question 13 I can do this question ☐

- Let $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \frac{1}{2}x & : x \in \mathbb{Q} \\ x & : x \notin \mathbb{Q} \end{cases}.$$

Is f differentiable at $c = 0$?

Question 14 I can do this question ☐

- Let $f : D \to \mathbb{R}$ and f be differentiable on some interval $I \subset D$. Prove that if $f'(x) \neq 0$ for each $x \in I$, then f is injective on I. Provide a counterexample to demonstrate the converse is not true.

Question 15 I can do this question ☐

- Consider the power series $\sum_{k=0}^{\infty} a_k x^k$ such that there exists an $m > 0$ and $M > 0$ with $0 < m \leq |a_k| < M$ for each $k \in \mathbb{N} \cup \{0\}$. Prove that the radius of convergence of the power series is 1.

Extension Qs

These questions are optional extension problems so you can gain more practice and confidence with these topics.

Question 16 I can do this question ☐

- (a) (L'Hospital's Rule ∞/∞) Let $f : (a, b) \to \mathbb{R}$ and $g : (a, b) \to \mathbb{R}$ be differentiable on (a, b) such that $g'(x) \neq 0$ for each $x \in (a, b)$. Prove that if $\lim_{x\to a}(g(x)) = \infty$ or $\lim_{x\to a}(g(x)) = -\infty$, then

$$\lim_{x\to a}\left(\frac{f'(x)}{g'(x)}\right) = L \Rightarrow \lim_{x\to a}\left(\frac{f(x)}{g(x)}\right) = L.$$

  (b) (L'Hospital's Rule 0/0) Let $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = g(a) = 0$, and $g'(x) \neq 0$ for all $x \in (a, b)$, then

$$\lim_{x\to a}\left(\frac{f'(x)}{g'(x)}\right) = L \Rightarrow \lim_{x\to a}\left(\frac{f(x)}{g(x)}\right) = L.$$

End Material

Review and Summary

Make sure you have fully understood and read the notes. Once the solutions for this worksheet have been released, check your answers and make a note of areas of concern. If you are still unsure then email your lecturer or drop by in office hours and we will be happy to help!

Completing Worksheets: Tips

- Proceed carefully and systematically from beginning to end making sure to note theorems and properties utilised.
- Work together with other students, discuss ideas and concepts.
- Prioritise carefulness over speed.
- Include enough information in your answers that you can easily understand your thinking when you come back to review the worksheets at a later date.
- Do ask for help if you get stuck!